

Md. Tanvir Sheikh

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Professional Summary

Machine Learning Engineer with hands-on experience building real-world AI systems, deep learning models, and end-to-end ML pipelines. Built production-ready applications using Python, PyTorch, and Streamlit, integrated with cloud backends like Supabase. Strong grasp of model training, evaluation, and deployment. Ready to solve real business problems with data-driven solutions.

Education

Alhaj Mockbul Hossain University College, Dhaka *BSc in Computer Science and Engineering*

April 2022 – Rinning

- **Coursework:** Object-Oriented Programming, Data Structures and Algorithms, Database Management System (DBMS), Computer Networks, Computer Architecture, Software Development Engineering, Information System Design, Artificial Intelligence, Machine Learning, etc.

Skills

Languages: Python

Frameworks: PyTorch, Scikit-learn, LangGraph, LangChain, NumPy, Pandas, Matplotlib, Seaborn, FastAPI, Streamlit, HuggingFace, Pydantic, Pillow

Tools: Git, Docker

ML Tools: XGBoost, LightGBM, CatBoost, dlib, Resemblyzer, Librosa, web rtcvad, ResNet-18

Database: MySQL, PostgreSQL, SQLite

Projects

Personal AI Assistant with RAG — LLM-Powered Chatbot

[Github](#) – [Live Demo](#)

- Built an end-to-end conversational AI assistant using LangGraph and Groq's Llama 3.1 for low-latency, streaming inference, enhanced with a RAG pipeline (query expansion, MMR search, ChromaDB, HuggingFace embeddings) for context-aware responses. Designed a multi-thread chat system with SQLite-based checkpointing for session persistence, wrapped in a custom Streamlit UI.
- **Tech Stack:** Python, LangGraph, LangChain, Groq ChromaDB, HuggingFace, Streamlit, SQLite

Attendance by Selfie (CNN)

[Github](#) – [Live Demo](#)

- A CNN-based attendance management system enabling teachers to mark attendance via classroom photo capture or voice recording of students. The platform features role-based access with separate student and teacher portals, facial recognition powered by dlib, speaker verification via Resemblyzer, cloud database storage with Supabase, secure password hashing with Bcrypt, and QR code generation using Segno.
- **Tech Stack:** Python, dlib, Resemblyzer, Supabase, Bcrypt, Segno

F1 Pit Stop Prediction — Kaggle Competition

[Kaggle](#)

- A multi-model ensemble pipeline built for predicting F1 pit stop outcomes, combining LightGBM, XGBoost, CatBoost, and a PyTorch ANN into a single optimized system. Applied GroupKFold cross-validation to prevent data leakage across race groups and engineered F1-domain-specific features to improve predictive accuracy. Explored advanced ensemble strategies including harmonic blending, rank averaging, and weighted stacking to maximize final performance.
- **Tech Stack:** Python, LightGBM, XGBoost, CatBoost, PyTorch, GroupKFold

Facial Emotion Detector (CNN — FER-2013)

[Github](#)

- A deep learning system that classifies 7 human facial emotions in real time. Built using both a custom CNN architecture and ResNet-18 transfer learning with PyTorch, trained on the FER-2013 dataset to compare performance between the two approaches.
- **Tech Stack:** Python, PyTorch, CNN, ResNet-18, FER-2013

Digit Classifier — From Scratch

[Github](#)

- A neural network built entirely from scratch without any deep learning frameworks, implementing forward propagation and backpropagation manually using only NumPy, Pandas, and math — demonstrating a solid foundational understanding of how neural networks work under the hood.
- **Tech Stack:** Python, NumPy